

ECON 120. Principles of Microeconomics

Behind the Supply Curve: Inputs & Costs

WHAT YOU WILL LEARN IN THIS CHAPTER

- What is the firm's production function?
- Why is production often subject to diminishing returns to inputs?
- What types of costs does a firm face, and how does the firm generate its marginal and average cost curves?
- Why do the firm's costs differ in the short run and in the long run?
- What is increasing returns to scale, and what advantage does it give?

THE PRODUCTION FUNCTION

- For a firm to earn an economic profit, it must produce a product that consumers want. This product is the firm's **output**.
- **Production** is the process of turning inputs into outputs.
- A **production function** is the relationship between the quantity of inputs a firm uses and the quantity of output it produces.

INPUTS

- Inputs are of two types:
 - A **fixed input** is an input whose quantity is fixed for a period of time and cannot be varied. Ex. Land, machinery, etc.
 - A **variable input** is an input whose quantity the firm can vary at any time. Ex. Labor, electricity, etc.
- The **long run** is the period in which all inputs can be varied.
- The **short run** is the period in which at least one input is fixed.

LEARN BY DOING: PRACTICE QUESTION 1

You own a deli. **Which of the following is most likely a fixed input at your deli?**

- a) the dining room
- b) the bread used to make sandwiches
- c) the tomato base used to make soups
- d) the employees

HOW MUCH SHOULD A FIRM PRODUCE?

Let's say you open a McDonald's franchise outlet...

- What are your fixed and variable costs?
- What are the immediate decisions you need to make?
 - How many workers to hire?
 - How to manage the inventory of supplies?

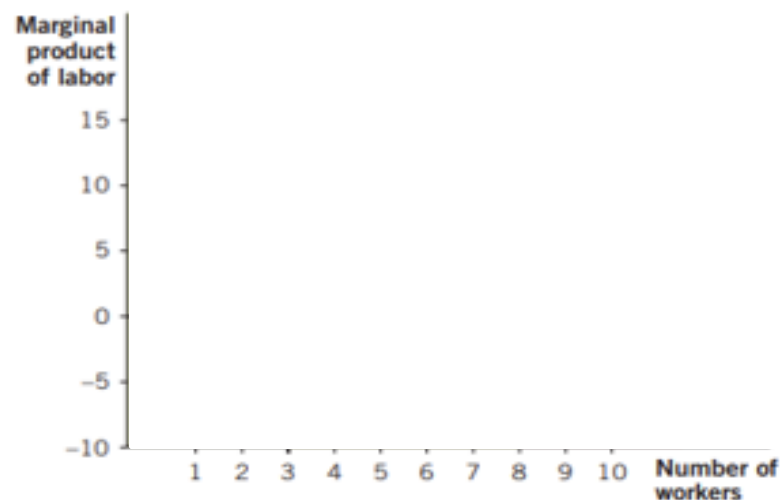
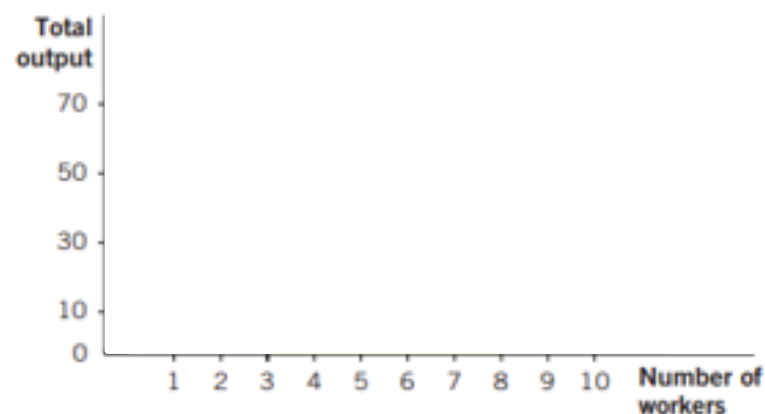


THE PRODUCTION FUNCTION AND MARGINAL PRODUCT

Suppose the production function for your outlet looks as follows:

Number of Workers	Total Output (No. of Meals/Hr.)	Marginal Product
0	0	-
1	5	
2	15	
3	30	
4	42	
5	52	
6	60	
7	65	
8	67	
9	54	
10	56	

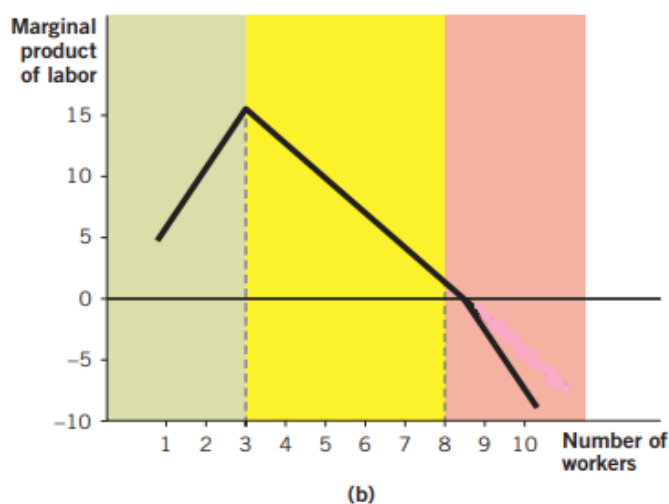
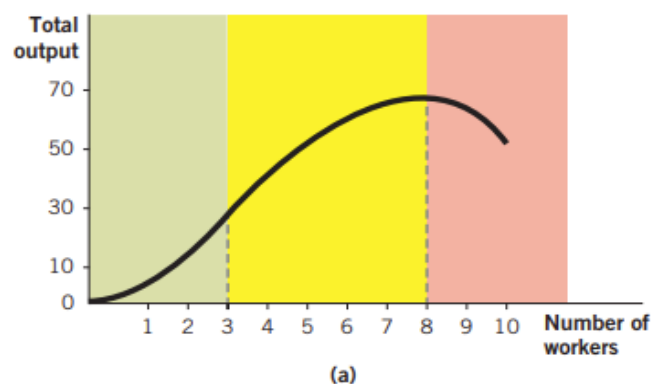
$$\text{Marginal Product} = \frac{\text{Change in Total Output}}{\text{Change in Number of Workers}}$$



THE PRODUCTION FUNCTION AND MARGINAL PRODUCT

Suppose the production function for your outlet looks as follows:

Number of workers	Total output (number of meals served per hour)	Marginal product of labor
0	0	5
1	5	10
2	15	15
3	30	12
4	42	10
5	52	8
6	60	5
7	65	2
8	67	-3
9	64	-8
10	56	



(a) **Total output** rises rapidly in the green zone from zero to three workers, rises less rapidly in the yellow zone between three and eight workers, and falls in the red zone after eight workers.

(b) The **marginal product of labor** rises in the green zone from zero to three workers, falls in the yellow zone from three to eight workers but remains positive, and becomes negative after eight workers.

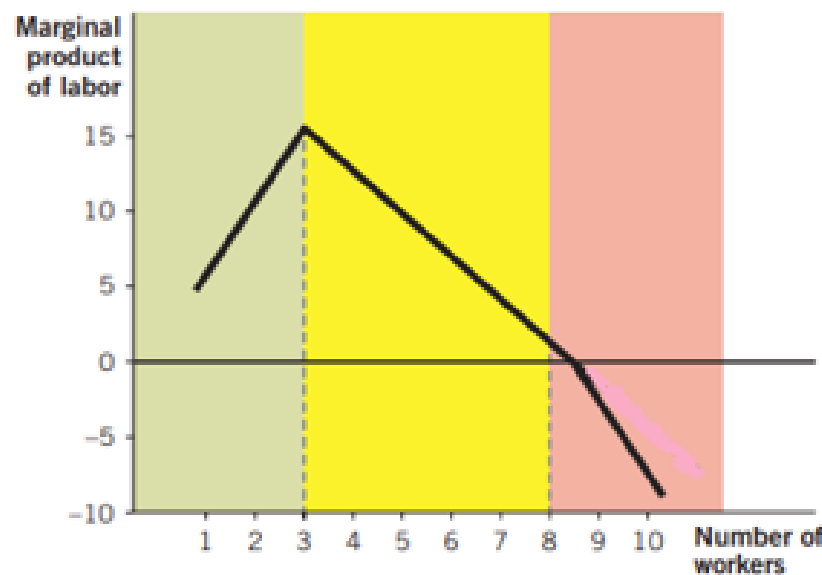
Source: Mateer and Coppock, Principles of Microeconomics, 4e.

TOTAL PRODUCT AND MARGINAL PRODUCT

- The **total product curve** shows how the quantity of output depends on the quantity of the variable input for a given quantity of the fixed input.
- The **marginal product** of an input is the additional quantity of output that is produced by using one more unit of that input.
- **Marginal product of labor (MPL)** is the change in output resulting from a one-unit increase in the amount of labor input ($\Delta Q/\Delta L$)
- MPL equals the slope of the total product curve.

DIMINISHING MARGINAL PRODUCT

- Going back to the marginal product of McDonald's workers, it rose from 5 to 15 meals per hour when we went from the first to the third worker.
- But the fourth worker only produced 12 extra meals. So, the marginal product began to decline.
- The point at which successive increases in inputs are associated with a slower rise in output is known as the point of **diminishing marginal product**.
- In which region of MPL would you like to operate?



LEARN BY DOING: PRACTICE QUESTION 2

- If one worker makes 14 basketballs,
 - two workers make 34 basketballs,
 - three workers make 45 basketballs, and
 - four workers make 50 basketballs, **which worker yielded the highest marginal product?**
-
- a) the first worker
 - b) the second worker
 - c) the third worker
 - d) the fourth worker

FROM THE PRODUCTION FUNCTION TO COST CURVES

- Until now, we have simply considered how much output each additional worker produces.
- But to decide how many workers to hire, we need to weigh the marginal benefit against the marginal cost of hiring each worker.
- There are two types of costs in the short run:
 - A **fixed cost** is a cost that is unavoidable and does **not** depend on the quantity of output produced. **It is the cost of the fixed input.**
 - A **variable cost** is a cost that depends on the quantity of output produced. **It is the cost of the variable input.**



"Sorry about this but we've outsourced production to Bangalore."

TOTAL COST CURVE

- The **total cost** of producing a given quantity of output is the sum of the fixed cost and the variable cost of producing that quantity of output.

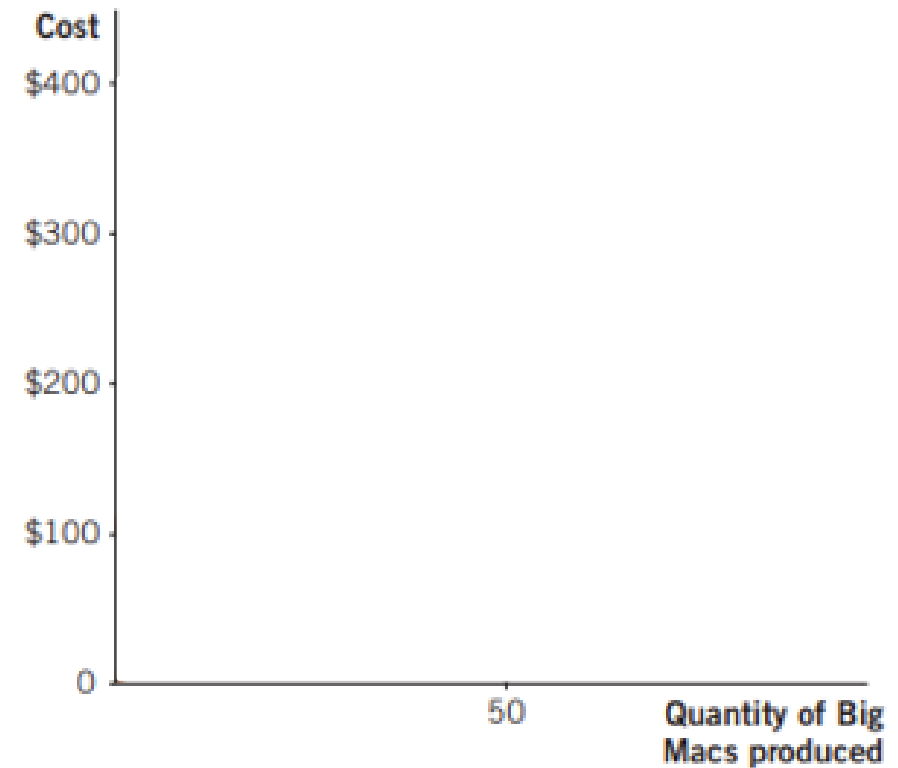
$$TC = FC + VC$$

- The **total cost curve** shows how total cost depends on the quantity of output.
- The total cost curve becomes **steeper as more output is produced**, a result of **diminishing returns**.
- With diminishing returns, additional units of output require more and more labor; therefore the cost increases.

TOTAL COST CURVE

Suppose the costs for your outlet looks as follows:

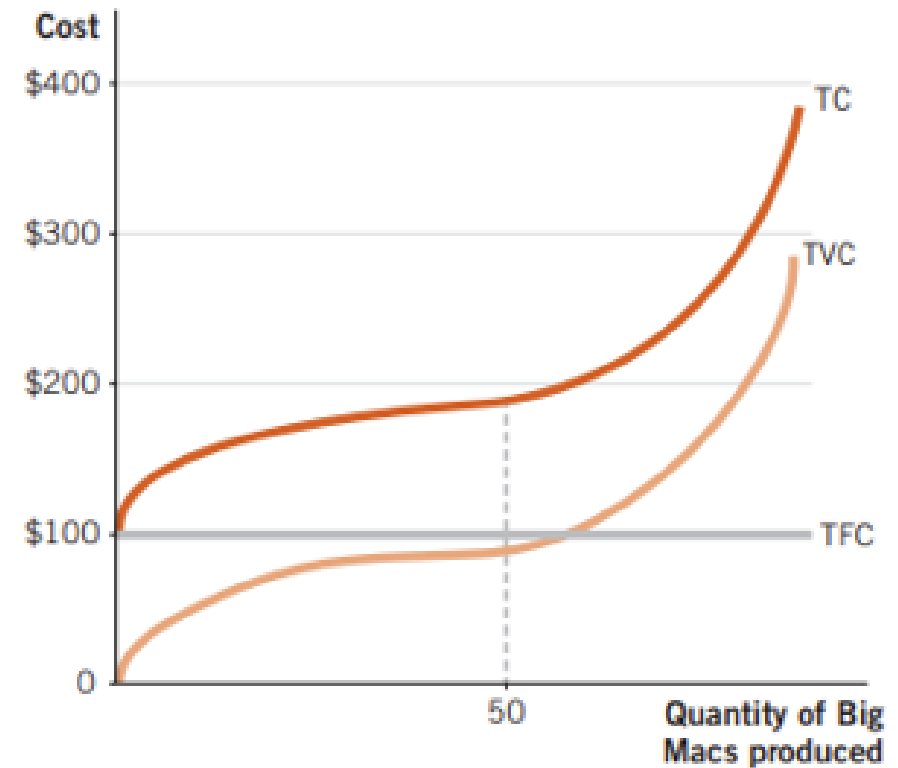
Quantity of Big Macs /Hr	Total Variable Cost	Total Fixed Cost	Total Cost
0	0	100	
10	30	100	
20	50	100	
30	65	100	
40	77	100	
50	87	100	
60	100	100	
70	120	100	
80	160	100	
90	220	100	
100	300	100	



TOTAL COST CURVE

Suppose the costs for your outlet looks as follows:

Quantity of Big Macs /Hr	Total Variable Cost	Total Fixed Cost	Total Cost
0	0	100	100
10	30	100	130
20	50	100	150
30	65	100	165
40	77	100	177
50	87	100	187
60	100	100	200
70	120	100	220
80	160	100	260
90	220	100	320
100	300	100	400



MARGINAL COST

The **marginal cost** is the change in total cost generated by one additional unit of output.

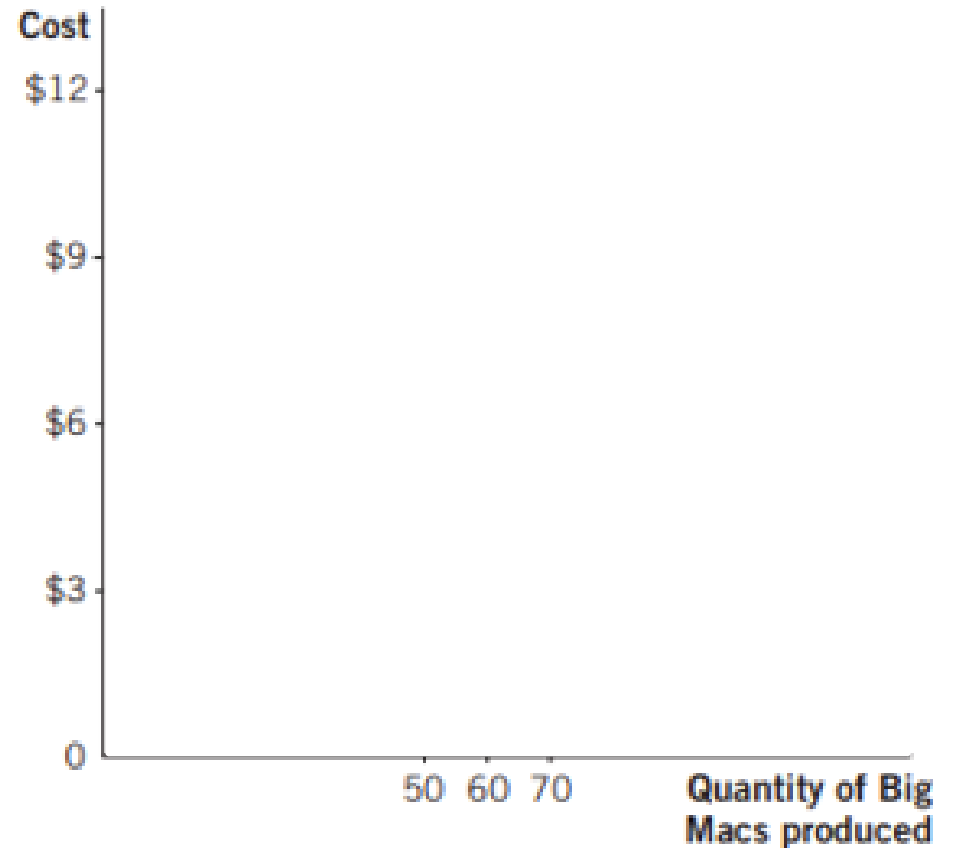
$$MC = \Delta TC / \Delta Q$$

where Δ = change, TC = total cost, and Q = quantity of output

MARGINAL CURVE

Suppose the costs for your outlet looks as follows:

Quantity of Big Macs /Hr	TVC	TFC	TC	Marginal Cost
0	0	100	100	
10	30	100	130	
20	50	100	150	
30	65	100	165	
40	77	100	177	
50	87	100	187	
60	100	100	200	
70	120	100	220	
80	160	100	260	
90	220	100	320	
100	300	100	400	



AVERAGE COST

Average total cost (often referred to simply as average cost) = total cost per unit of output produced.

$$ATC = TC/Q$$

Average fixed cost = fixed cost per unit of output produced.

$$AFC = FC/Q$$

Average variable cost = variable cost per unit of output produced.

$$AVC = VC/Q$$

COSTS IN THE SHORT RUN

Suppose you only produce Bic Macs, and these are the costs of production in the short run.

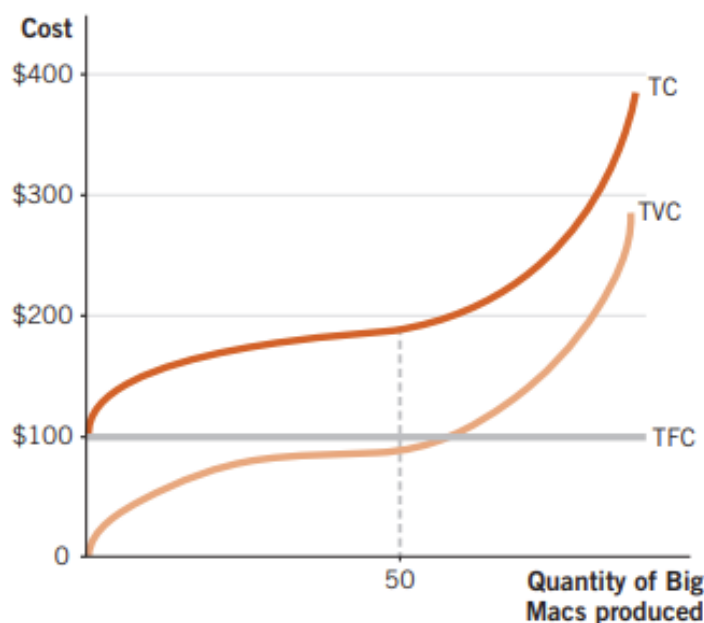
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Quantity (Q = Big Macs produced per hour)	Total variable cost	Total fixed cost	Total cost	Average variable cost	Average fixed cost	Average total cost	Marginal cost
Abbreviation:	TVC	TFC	TC	AVC	AFC	ATC	MC
Formula:			$TVC + TFC$	$TVC \div Q$	$TFC \div Q$	$TC \div Q$ or $AVC + AFC$	$\Delta TVC \div \Delta Q$ or $\Delta TC \div \Delta Q$
0	\$0.00	\$100.00	\$100.00				
10	30.00	100.00	130.00	\$3.00	\$10.00	\$13.00	\$3.00
20	50.00	100.00	150.00	2.50	5.00	7.50	2.00
30	65.00	100.00	165.00	2.17	3.33	5.50	1.50
40	77.00	100.00	177.00	1.93	2.50	4.43	1.20
50	87.00	100.00	187.00	1.74	2.00	3.74	1.00
60	100.00	100.00	200.00	1.67	1.67	3.34	1.30
70	120.00	100.00	220.00	1.71	1.43	3.14	2.00
80	160.00	100.00	260.00	2.00	1.25	3.25	4.00
90	220.00	100.00	320.00	2.44	1.11	3.55	6.00
100	300.00	100.00	400.00	3.00	1.00	4.00	8.00

Source: Mateer and Coppock, Principles of Microeconomics, 4e.

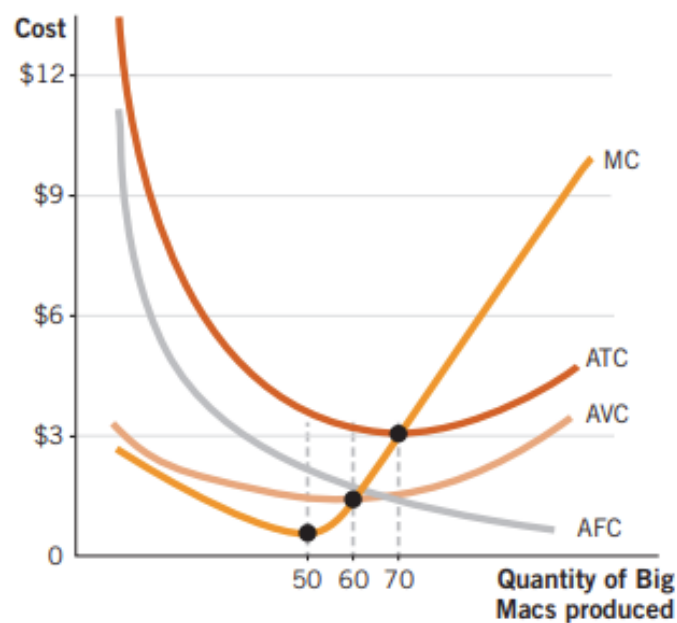
COST CURVES IN THE SHORT RUN

The table is visualized below.

- (a) The total variable cost (TVC) dictates the shape of the total cost (TC) curve. After 50 Big Macs, diminishing marginal product causes the total cost curve to rise at an increasing rate. Notice that the total fixed cost curve (TFC) stays constant, or flat.
- (b) The marginal cost curve (MC) reaches its minimum before average variable cost (AVC) and average total cost (ATC). Marginals always lead the average variable and average total costs either up or down. Average fixed cost (AFC), which has no variable component, continues to fall with increased quantity, because total fixed costs are spread across more units.



(a) Total Costs



(b) Average and Marginal Costs

Source: Mateer and Coppock, Principles of Microeconomics, 4e.

LEARN BY DOING: PRACTICE QUESTION 3

You like to bake cakes, which you sell in your neighborhood. Currently you can bake 4 cakes at a total cost of \$40.

Suppose you could produce one more cake (the fifth) at a marginal cost of \$5.

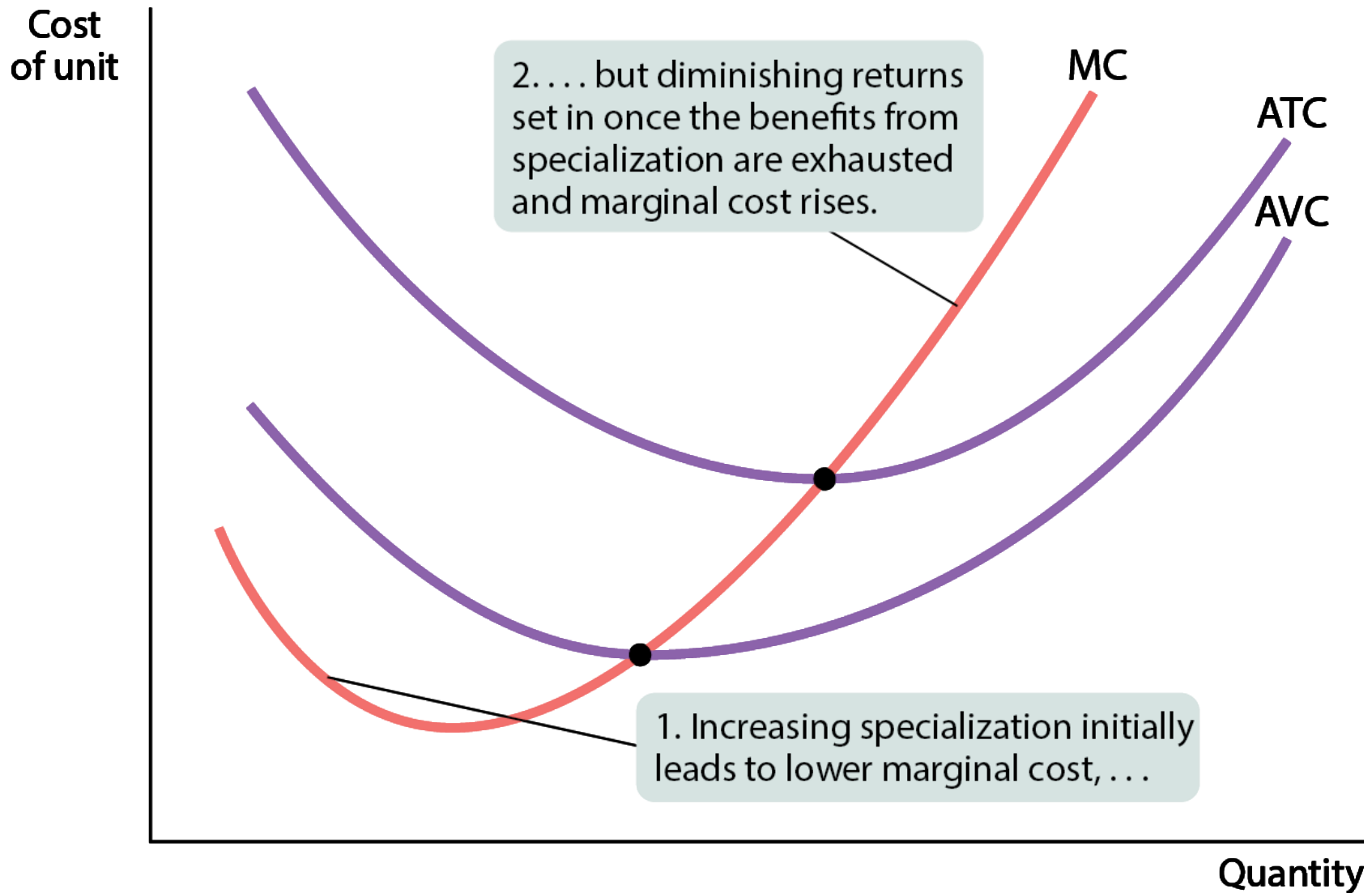
If you do produce that fifth cake, what will your average total cost be? Has your average total cost increased or decreased?

- a) Your average total cost has decreased to \$11.
- b) Your average total cost has decreased to \$9.
- c) Your average total cost has increased to \$9.
- d) Your average total cost has increased to \$11.

MARGINAL COST CURVE

- Focus on the upward-sloping part of the marginal cost curve. (No firm will operate on the downward-sloping part because it can produce more and increase profits as costs are falling.)
- So, the marginal cost curve is upward-sloping. Why?
- **Because in this portion there are diminishing returns to inputs.** As output increases, the marginal product of labor is declining (the yellow portion of the MPL graph).
- This implies that **more and more of workers must be used to produce each additional unit of output** as the amount of output rises.
- And since each unit of the variable input must be paid the same wage, the **cost per additional unit of output also rises.** Therefore, the *marginal* cost of each Big Mac is rising.

MARGINAL COST CURVE



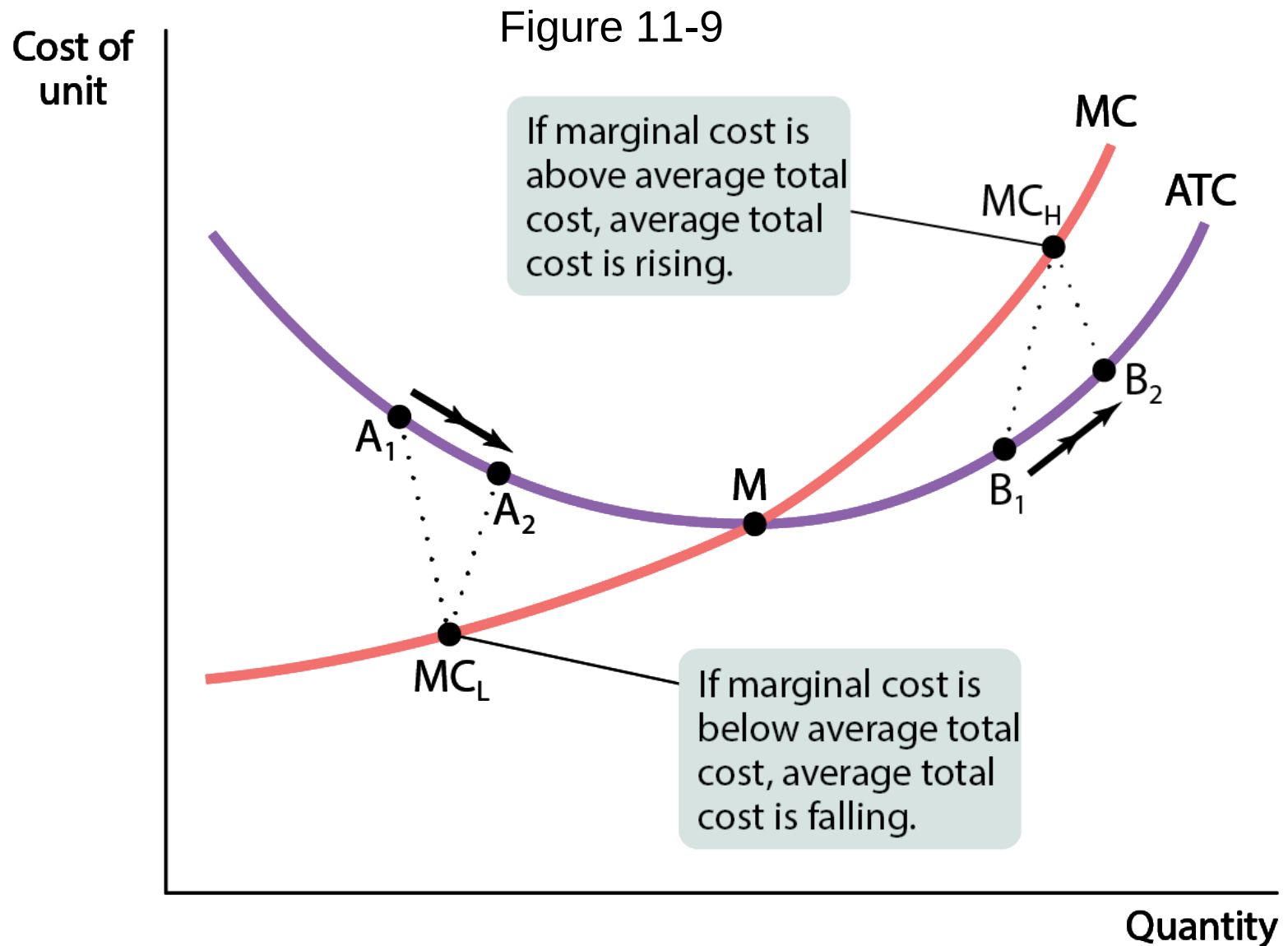
AVERAGE TOTAL COST CURVE

- The average total cost curve is U-shaped. Why?
- **Increasing output has two opposing effects on average total cost:**
 - **The spreading effect:** The larger the output, the more output over which fixed cost is spread, leading to lower average fixed cost.
 - **The diminishing returns effect:** The larger the output, the more variable input required to produce additional units, which leads to higher average variable cost.
- In the downward-sloping portion of the ATC, the spreading effect dominates. In the upward-sloping portion, the diminishing returns effect dominates.

MINIMUM AVERAGE TOTAL COST

- **The minimum-cost output** is the quantity of output at which average total cost is lowest—the bottom of the U-shaped average total cost curve.
- **Three general principles are always true about a firm's marginal cost and average total cost curves:**
 1. **At the minimum-cost output**, average total cost is equal to marginal cost.
 2. **At output less than the minimum-cost output**, marginal cost is less than average total cost and average total cost is falling.
 3. **At output greater than the minimum-cost output**, marginal cost is greater than average total cost and average total cost is rising.

THE RELATIONSHIP BETWEEN THE AVERAGE TOTAL COST AND THE MARGINAL COST CURVES



PUTTING THE FOUR COST CURVES TOGETHER

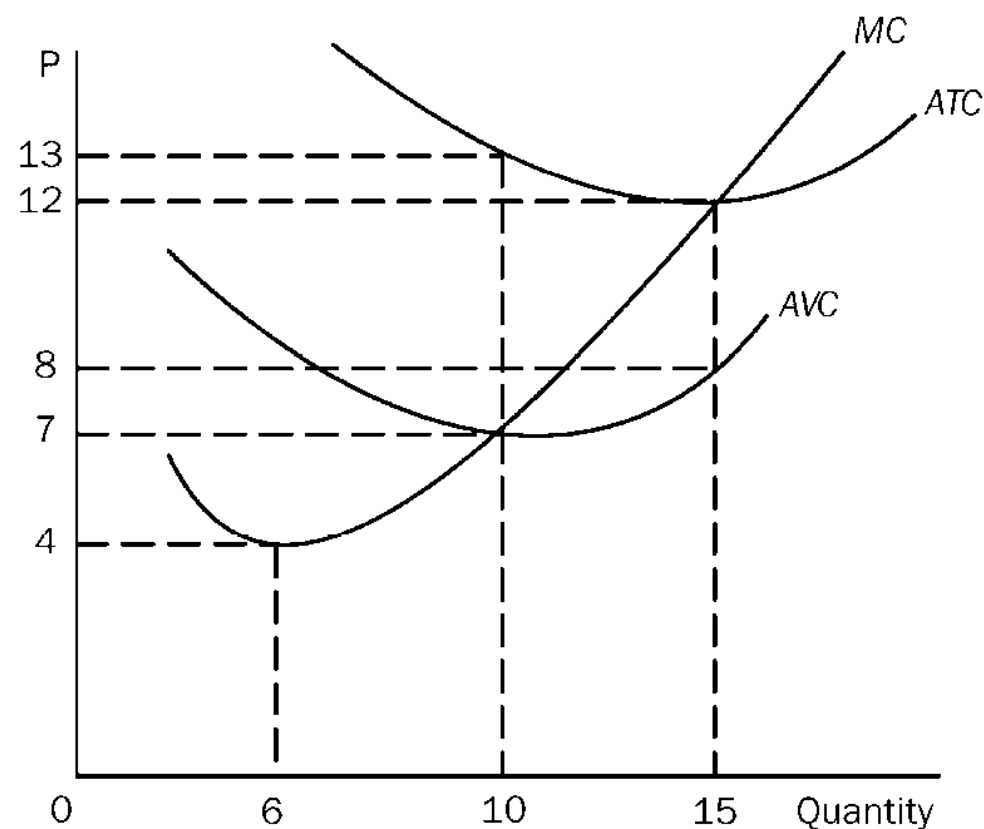
1. Marginal cost slopes upward because of **diminishing returns**.
2. Average variable cost also slopes upward but is flatter than the marginal cost curve.
3. Average fixed cost slopes downward because of the **spreading effect**.
4. The marginal cost curve intersects the average total cost curve from below, crossing it at its lowest point.

LEARN BY DOING: PRACTICE QUESTION 4a

Consider the cost curves for one very small firm in a large market.

If the firm produces 10 units of output, its average total cost is

- a) 6
- b) 7
- c) 12
- d) 13

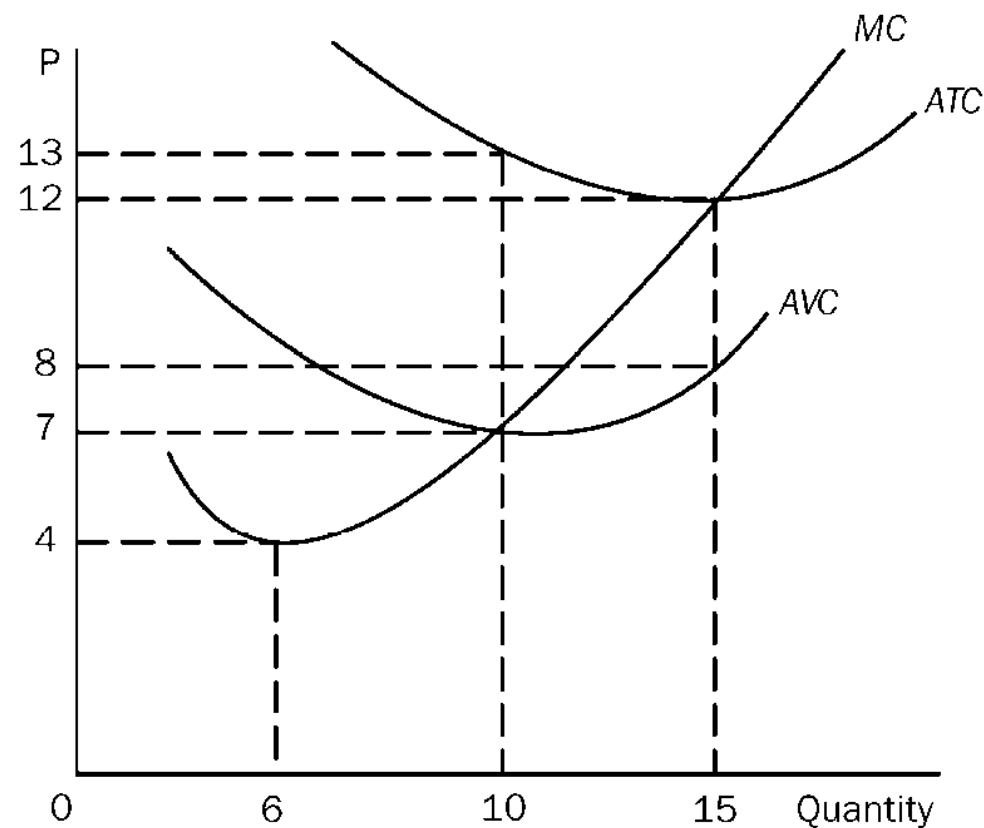


LEARN BY DOING: PRACTICE QUESTION 4b

Consider the cost curves for one very small firm in a large market.

If the firm produces 15 units of output, its average fixed cost is

- a) 4
- b) 5
- c) 6
- d) 60

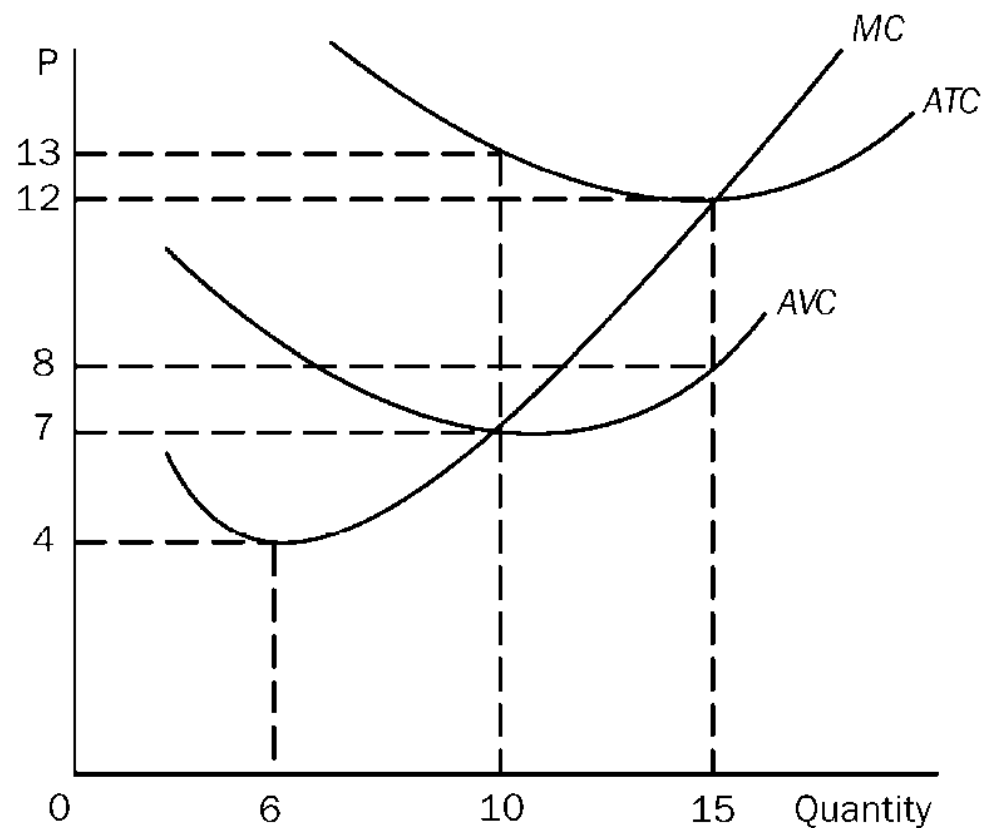


LEARN BY DOING: PRACTICE QUESTION 4c

Consider the cost curves for one very small firm in a large market.

If the firm produces 10 units of output, its total cost is

- a) 7
- b) 13
- c) 70
- d) 130

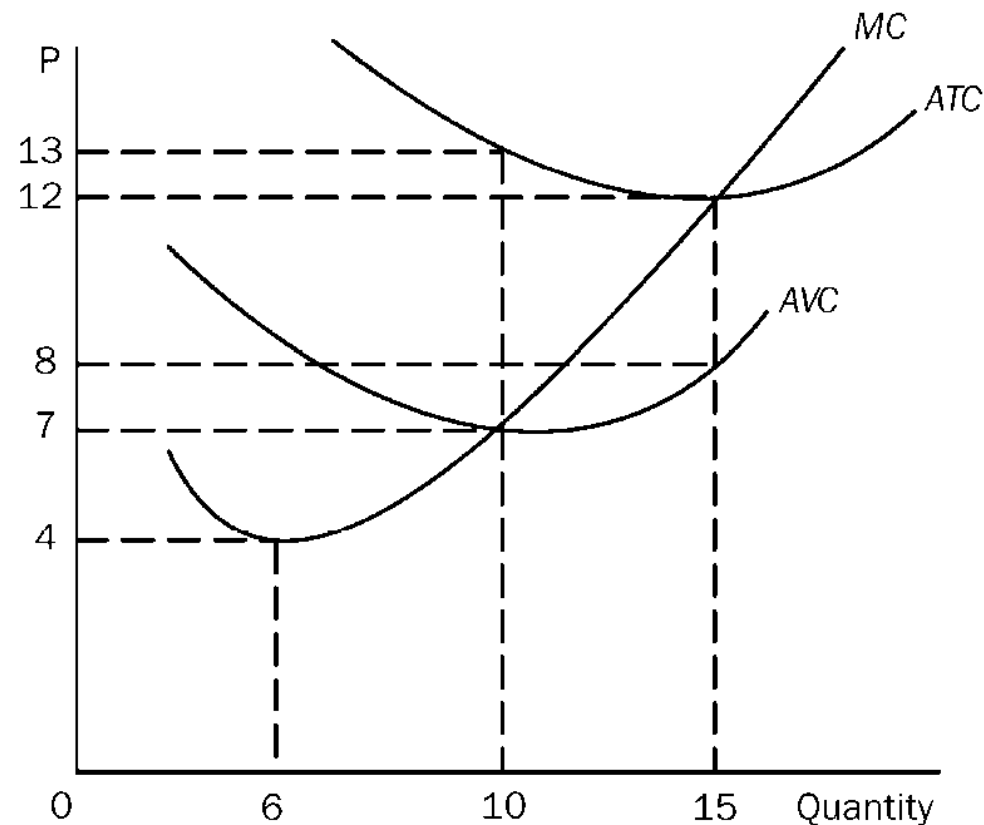


LEARN BY DOING: PRACTICE QUESTION 4d

Consider the cost curves for one very small firm in a large market.

Diminishing returns to the variable factor of production for this firm set in at

- a) $Q=6$
- b) $Q=10$
- c) $Q=15$
- d) None of the Above

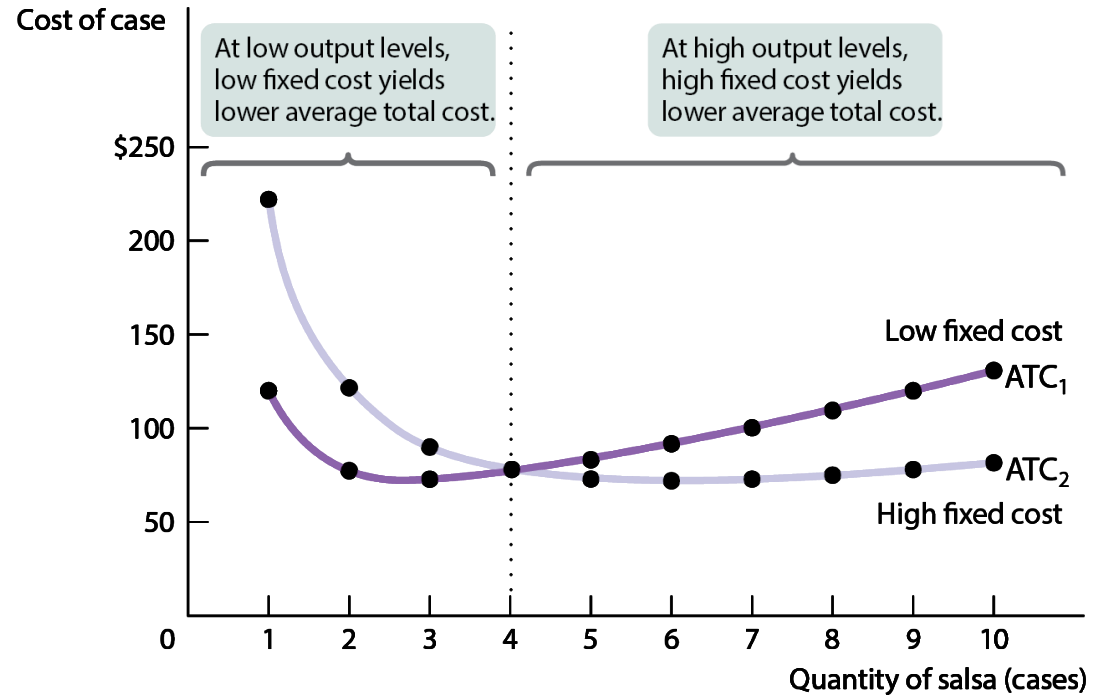


COSTS IN THE LONG RUN

- **All inputs are variable in the long run.** This means that in the long run, fixed cost (like factory size) may also vary.
- One way in which firms can adjust in the long run is by changing the **scale**, or size, of the production process. If the business is expected to grow, the firm can scale up production. If the business is faltering, it can scale back its operations.
- Economists refer to the quantity of output that minimizes the average total cost in the long run as the **efficient scale**.
- **The firm will choose its fixed cost in the long run based on the level of output it expects to produce.**

CHOOSING THE LEVEL OF FIXED COST

- Suppose a salsa shop is considering buying new equipment. It will affect cost in two ways:
 - Higher fixed cost in the short run
 - Lower variable cost because workers are now more productive.
- There is a trade-off between higher fixed cost and lower variable cost for any given output level, and vice versa.
- ATC₂ is the case where the shop buys the new equipment.



Quantity of salsa (cases)	Low fixed cost (FC = \$108)			High fixed cost (FC = \$216)		
	High variable cost	Total cost	Average total cost of case ATC ₁	Low variable cost	Total cost	Average total cost of case ATC ₂
1	\$12	\$120	\$120.00	\$6	\$222	\$222.00
2	48	156	78.00	24	240	120.00
3	108	216	72.00	54	270	90.00
4	192	300	75.00	96	312	78.00
5	300	408	81.60	150	366	73.20
6	432	540	90.00	216	432	72.00
7	588	696	99.43	294	510	72.86
8	768	876	109.50	384	600	75.00
9	972	1,080	120.00	486	702	78.00
10	1,200	1,308	130.80	600	816	81.60

THE LONG-RUN AVERAGE TOTAL COST CURVE

- The **long-run average total cost curve** shows the relationship between output and average total cost when fixed cost has been chosen to minimize average total cost for each level of output.
 - (We assume the firm has chosen the cheapest plant size for each output level.)

SHORT-RUN AND LONG-RUN AVERAGE TOTAL COST CURVES

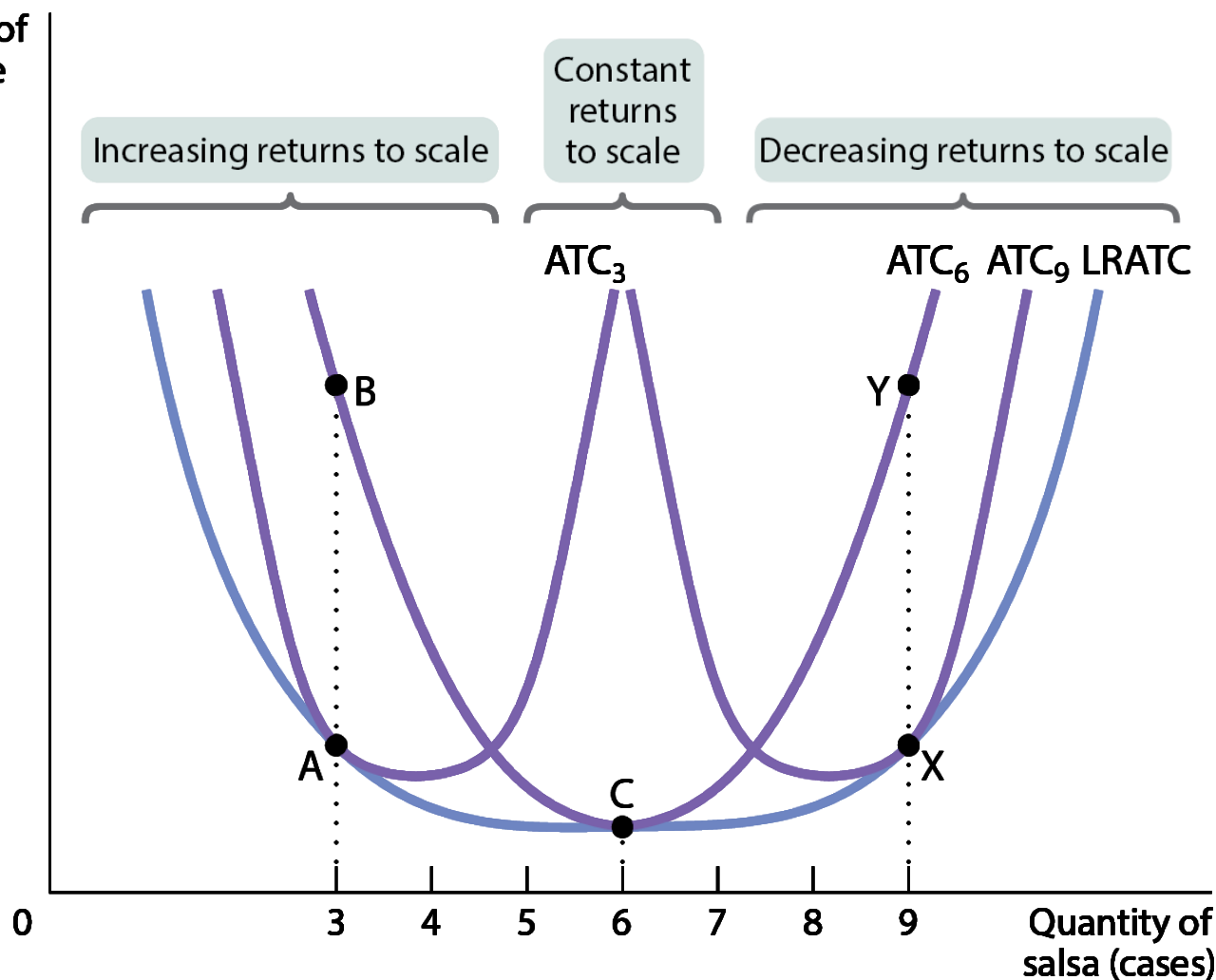
Figure 11-12

Cost of case

Increasing returns to scale

Constant returns to scale

Decreasing returns to scale



Short-run and long-run average total cost curves differ because a firm can choose its fixed cost in the long run.

If the firm chooses the fixed cost that minimizes short-run ATC at an output of 6, and produces 6, it's at point C.

If it produces only 3, it'll move to point B.

If the firm expects to produce 3 cases for a long time, it'll reduce its fixed cost and move to point A.

If it produces 9 (point Y) and expects to continue this for a long time, it'll increase its fixed cost and move to point X.

RETURNS TO SCALE

- There are **increasing returns to scale (economies of scale)** when long-run average total cost declines as output increases.

Ex. Supermarkets can benefit from economies of scale because they can buy food in bulk and get lower average costs. If you had a delivery of just 100 cartons of milk the average cost is quite high. The marginal cost of delivering 10,000 cartons is quite low. You still need to pay only one driver; the fuel costs will be similar.

- There are **decreasing returns to scale (diseconomies of scale)** when long-run average total cost increases as output increases.

Ex. Bigger isn't always better! Sometimes a company grows so large that coordination problems make costs rise. For this reason, a larger firm can become less effective at holding down long-run average total costs and experience diseconomies of scale.

- There are **constant returns to scale** when long-run average total cost is constant as output increases.